Spoken Tutorial – AI, ML, and Deep Learning: An Introduction

Title: AI, ML, and Deep Learning: An Introduction

Module	Introduction to AI Concepts
Episode	Understanding AI, Machine Learning, and Deep Learning
Learning Objective	At the end of this tutorial learner will be able to 1. Define Artificial Intelligence, Machine Learning, and Deep Learning. 2. Understand the relationship between AI, ML, and Deep Learning. 3. Differentiate between traditional programming and machine learning using an example. 4. Identify scenarios where Deep Learning is particularly effective.
Approx. Duration	3 minutes
Outline	• Introduction to AI • Understanding Machine Learning • The Analogy: Traditional Programming vs. Machine Learning • Exploring Deep Learning • Summary and Assignment
Meta Tags	AI, Machine Learning, Deep Learning, Artificial Intelligence, ML, DL, Spoken Tutorial, EduPyramids
Pre-requisite Tutorial	Basic computer literacy and a stable internet connection.

Script

Visual Cue	Narration
A title slide with 'AI, ML, and Deep Learning: An Introduction' prominently displayed, with abstract tech background.	Welcome to this Spoken Tutorial. This tutorial is on AI, ML, and Deep Learning: An Introduction.
A slide showing bullet points for learning objectives, with icons representing understanding and analysis.	In this tutorial, you will learn to. First, you will define Artificial Intelligence. You will also define Machine Learning and Deep Learning. Next, you will understand the relationship between AI, ML, and Deep Learning. Then, you will differentiate between traditional programming and machine learning. We will use an example to illustrate this. Finally, you will identify scenarios where Deep Learning is particularly effective.
An image showing a desktop computer, a laptop, and a mobile phone, all displaying a web browser interface.	Here, I am using a browser on a computer or mobile.

An icon representing basic computer literacy and a Wi-Fi symbol for internet connection.	To follow this tutorial, you must have basic computer literacy. You also need a stable internet connection. No coding skills are required for this introduction.
A futuristic robot thinking, surrounded by data and algorithms, representing human-like intelligence.	At its core, Artificial Intelligence, or AI, is a broad field of computer science. Its main goal is simple: to build smart machines. These machines perform tasks that usually need human intelligence.
Timeline showing milestones in AI history, from 1950s to present, with a thought bubble 'Can machines think?'	This could be anything from understanding human language. It also includes recognizing images and solving complex problems. The idea isn't new. It has been a dream of scientists since the 1950s. Pioneers asked, 'Can machines think?' AI is truly the entire universe of making computers intelligent.
A computer screen showing data points forming a pattern, with a machine learning model 'learning' from it, a brain icon.	Moving on, Machine Learning, or ML, is a specific type of AI. In traditional programming, we give a computer a detailed list of rules to follow. With Machine Learning, we give it a lot of data instead. We then let the machine learn the rules for itself. It's all about recognizing patterns from examples within that data.
A flow chart showing strict rules for identifying a cat: 'Has pointy ears? -> Yes -> Has whiskers? -> Yes -> Is it a cat?' with a picture of a cat.	Let's use an analogy to understand this better. Imagine you want to program a computer to identify a cat. With traditional programming, you would write specific rules. For example, you would define a cat by its pointy ears, whiskers, and fur. You would also define it by its four legs. Notice how this approach is fragile. What if the cat's ears are folded? Or what if it's partially hidden?
A collage of thousands of diverse cat pictures feeding into a machine learning model, with a final output 'Cat identified'.	Now, for Machine Learning, the approach is different. Instead, you show the computer thousands of pictures labeled 'cat.' The ML model analyzes all these images. It then learns the common patterns and features that define a cat on its own. As you can see, it builds its own internal 'rules'. These rules are much more flexible and accurate.
An abstract representation of a neural network with multiple interconnected layers, glowing nodes.	Finally, let's explore Deep Learning. Deep Learning is a super-powered, more advanced version of Machine Learning. It uses a special structure called a neural network. This network is inspired by the human brain.

A close-up of a deep neural network showing many layers of 'neurons' and connections, emphasizing depth.	These networks have many layers of 'neurons' stacked on top of each other. Because there are many layers, we call it 'deep.' This deep structure allows it to learn complex patterns from data. It can learn much more than traditional ML.
Collage of icons representing self-driving cars, voice assistants (like Siri/Alexa), and generative AI (art/text creation).	Deep learning is the magic behind many modern applications. For example, it powers self-driving cars that recognize pedestrians. It also powers voice assistants that understand your commands. And it drives the generative AI tools we use today. It enables machines to perform highly intricate tasks with impressive accuracy.
A Venn diagram showing AI as the largest circle, ML as a subset within AI, and Deep Learning as a subset within ML.	Let's summarize what we have learned. In this tutorial, we understood that AI is a broad field. Its goal is to make machines intelligent. ML is a type of AI where machines learn from data. Deep Learning is an advanced ML. It uses deep neural networks for complex pattern recognition.
A notepad and pen icon, next to various fruits (apple, banana, orange).	Now, for a small activity. Think of a simple task, like identifying different types of fruits. Briefly describe how you would approach this task. First, consider using traditional programming. Then, describe how you would approach it using Machine Learning. Notice the difference in approach for each method.
EduPyramids Educational Services Private Limited logo.	This brings us to the end of this tutorial. This Spoken Tutorial comes from EduPyramids Educational Services Private Limited. They are located at SINE, IIT Bombay.
A simple 'Thank You' message with a pleasant background.	Thank you for joining!